

Lower bounds on Locality Sensitive Hashing

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ABSTRACT

Given a metric space (X, d_X) , $c \geq 1$, $r > 0$, and $p, q \in [0, 1]$, a distribution over mappings $\mathcal{H} : X \rightarrow \mathbb{N}$ is called a (r, cr, p, q) -sensitive hash family if any two points in X at distance at most r are mapped by $\mathcal{H}$ to the same value with probability at least p , and any two points at distance greater than cr are mapped by $\mathcal{H}$ to the same value with probability at most q . This notion was introduced by Indyk and Motwani in 1998 as the basis for an efficient approximate nearest neighbor search algorithm, and has since been used extensively for this purpose. The performance of these algorithms is governed by the parameter $\rho = \frac{\log(1/p)}{\log(1/q)}$, and constructing hash families with small ρ automatically yields improved nearest neighbor algorithms. Here we show that for $X = \ell_1$ it is impossible to achieve $\rho \leq \frac{1}{2c}$. This almost matches the construction of Indyk and Motwani which achieves $\rho \leq \frac{1}{c}$.

Categories and Subject Descriptors

G.3 [Probability and Statistics]: Probabilistic algorithms;
E.1 [Data Structures]: Miscellaneous

General Terms

Algorithms, Theory

Keywords

Nearest Neighbor Search, Locality Sensitive Hashing, Lower Bounds

1. INTRODUCTION

In this note we study the complexity of finding the nearest neighbor of a query point in certain high dimensional spaces

^{*}Supported by NSF Grant ITR-0331640.

[†]Supported by NSF Grant ITR-0331640 and a Stanford Graduate Fellowship.

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SCG'06, June 5–7, 2006, Sedona, Arizona, USA.

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using *Locality Sensitive Hashing* (LSH). The nearest neighbor problem is formulated as follows: Given a database of n points in a metric space, preprocess it so that given a new query point it is possible to quickly find the point closest to it in the data set. This fundamental problem arises in numerous applications, including data mining, information retrieval, and image search, where distinctive features of the objects are represented as points in $\mathbb{R}^d$. There is a vast amount of literature on this topic, and we shall not attempt to discuss it here. We refer the interested reader to the papers [6, 5, 4, 7], and especially to the references therein, for background on the nearest neighbor problem.

While the exact nearest neighbor problem seems to suffer from the “curse of dimensionality”, many efficient techniques have been devised for finding an approximate solution whose distance from the query point is at most c times its distance from the nearest neighbor. One of the most versatile and efficient methods for approximate nearest neighbor search is based on Locality Sensitive Hashing, as introduced by Indyk and Motwani in 1998 [6]. This method has been refined and improved in several papers- the most recent algorithm can be found in [4]. We also refer the reader to the LSH website, where more information on this algorithm can be found, including its implementation and code- all this can be found at <http://web.mit.edu/andoni/www/LSH/index.html>. The LSH approach to the approximate nearest neighbor problem is based on the following concept.

DEFINITION 1.1. Let (X, d_X) be a metric space, $r, R > 0$ and $p, q \in [0, 1]$. A distribution over mappings $\mathcal{H} : X \rightarrow \mathbb{N}$ is called a (r, R, p, q) -sensitive hash family if for any $x, y \in X$,

- $d_X(x, y) \leq r \implies \Pr_{\mathcal{H}}[\mathcal{H}(x) = \mathcal{H}(y)] \geq p$.
- $d_X(x, y) > R \implies \Pr_{\mathcal{H}}[\mathcal{H}(x) = \mathcal{H}(y)] \leq q$.

Given $c \geq 1$ and $q \in (0, 1)$ we define $\rho_X(c, q)$ to be the smallest constant $\rho > 0$ such that for every $r > 0$ there exists $p \in (0, 1)$ and a (r, cr, p, q) -sensitive hash family $\mathcal{H} : X \rightarrow \mathbb{N}$ with $\frac{\log(1/p)}{\log(1/q)} \leq \rho$. In other words

$$\rho_X(c, q) = \sup_{r>0} \inf \left\{ \frac{\log(1/p)}{\log(1/q)} : \exists (r, cr, p, q) \text{-sensitive hash family } \mathcal{H} : X \rightarrow \mathbb{N} \right\} . \quad (1)$$

Of particular interest is the case $X = \ell_s^d$, for some $s > 0$ and $d \in \mathbb{N}$. Here, and in what follows, ℓ_s^d denotes the space $\mathbb{R}^d$ equipped with the ℓ_s norm

$$\|(x_1, \dots, x_d)\|_s = (|x_1|^s + \dots + |x_d|^s)^{1/s}$$

(this is only a quasi-norm when $0 < s < 1$). In this case we define

$$\rho_s(c) = \sup_{0 < q < 1} \limsup_{d \rightarrow \infty} \rho_{\ell_s^d}(c, q) .$$

The importance of these parameters stems from the following application to approximate nearest neighbor search. It will be convenient to discuss it in the framework of the following decision version of the c -approximate nearest neighbor problem: Given a query point, find any element of the data set which is at distance at most cr from it, provided that there is a data point at distance at most r from the query point. This decision version is known as the (r, cr) -near neighbor problem. It is well known that the reduction to the decision version adds only a logarithmic factor in the time and space complexity [6, 5]. The following theorem was proved in [6]; the exact formulation presented here is taken from [4].

THEOREM 1.2. *Let (X, d_X) be a metric on a subset of $\mathbb{R}^d$. Suppose that (X, d_X) admits a (r, cr, p, q) -sensitive hash family $\mathcal{H}$, and write $\rho = \frac{\log(1/p)}{\log(1/q)}$. Then for any $n \geq \frac{1}{q}$ there exists a randomized algorithm for (r, cr) -near neighbor on n -point subsets of X which uses $O(dn + n^{1+\rho})$ space, with query time dominated by $O(n^\rho)$ distance computations and $O(n^\rho \log_{1/q} n)$ evaluations of hash functions from $\mathcal{H}$.*

Thus, obtaining bounds on $\rho_X(c)$ is of great algorithmic interest. It is proved in [6] that $\rho_1(c) \leq 1/c$, and for small values of c , namely $c \in [1, 10]$, it was shown in [4] that this inequality is strict. We refer to [4] for numerical data on the best known estimates for $\rho_1(c)$ for small c . For $s = 2$ a recent result of Andoni and Indyk [1] shows that $\rho_2(c) \leq 1/c^2$, and for general $s \in (0, 2]$ the best known bounds [4] are $\rho_s(c) \leq \max\{1/c, 1/c^s\}$.

The main purpose of this note is to obtain lower bounds on $\rho_1(c)$ and $\rho_2(c)$ which nearly match the bounds obtained from the constructions in [6, 4, 1]. Our main result is:

THEOREM 1.3. *For every $c, s \geq 1$,*

$$\rho_s(c) \geq \frac{e^{\frac{1}{c^s}} - 1}{e^{\frac{1}{c^s}} + 1} \geq \frac{e - 1}{e + 1} \cdot \frac{1}{c^s} \geq \frac{0.462}{c^s} . \quad (2)$$

The second to last inequality in (2) follows from concavity of the function $t \mapsto \frac{e^t - 1}{e^t + 1}$ on $[0, \infty)$. Observe also that as $c \rightarrow \infty$, $\frac{e^{1/c^s} - 1}{e^{1/c^s} + 1} \sim \frac{1}{2c^s}$. It would be very interesting to determine $\limsup_{c \rightarrow \infty} c \cdot \rho_1(c)$ exactly- due to Theorem 1.3 and the results of [6] we currently know that this number is in the interval $[1/2, 1]$.

2. PROOF OF THEOREM 1.2

The basic idea in the proof of Theorem 1.3 is simple. Choose a random point $x \in \{0, 1\}^d$ and consider the random subset A of the cube $\{0, 1\}^d$ consisting of points u for which $\mathcal{H}(u) = \mathcal{H}(x)$. The second condition in Definition 1.1 forces A to be small in expectation. But, when A is small we can bound from above the probability that after r steps, the random walk starting at a random point in A will end up in A . We obtain this upper bound using a Fourier analytic argument, and in combination with the first condition in Definition 1.1 we deduce the desired bound on $\rho_1(c)$.

Theorem 1.3 follows from the following result:

PROPOSITION 2.1. *Let $\mathcal{H}$ be a (r, R, p, q) -sensitive hash family on the Hamming cube $(\{0, 1\}^d, \|\cdot\|_1)$. Assume that r is an odd integer and that $R < \frac{d}{2}$. Then*

$$p \leq \left(q + e^{-\frac{1}{d}(\frac{d}{2} - R)^2} \right)^{\frac{e^{2r/d} - 1}{e^{2r/d} + 1}} .$$

Choosing $R \approx \frac{d}{2} - \sqrt{d \log d}$ and $r \approx R/c$ in Proposition 2.1, and letting $d \rightarrow \infty$, yields Theorem 1.3 in the case $s = 1$. The case of general $s \geq 1$ follows from the fact that for $x, y \in \{0, 1\}^d$, $\|x - y\|_s = \|x - y\|_1^{1/s}$.

REMARK 2.1. Proposition 2.1 implies a non-trivial lower bound on $\frac{\log(1/p)}{\log(1/q)}$ for any (r, cr, p, q) -sensitive hash family on $(\{0, 1\}^d, \|\cdot\|_1)$ even if q is allowed to depend on d . Observe that with the definition given in (1), Theorem 1.3 implies such a lower bound only for constant q . But, Proposition 2.1 is much stronger, and implies a bound which asymptotically coincides with the lower bound in 1.3 for every $q \geq 2^{-o(d)}$.

The proof of Proposition 2.1 will be broken into a few lemmas.

LEMMA 2.2. *Let $\mathcal{H}$ be a (r, R, p, q) -sensitive hash family on the Hamming cube $(\{0, 1\}^d, \|\cdot\|_1)$, and fix $x \in \{0, 1\}^d$. Then*

$$\mathbb{E} |\mathcal{H}^{-1}(\mathcal{H}(x))| \leq \sum_{k=0}^{\lfloor R \rfloor} \binom{d}{k} + q \cdot \sum_{k=\lfloor R \rfloor+1}^d \binom{d}{k} .$$

PROOF. We simply write

$$\begin{aligned} \mathbb{E} |\mathcal{H}^{-1}(\mathcal{H}(x))| &= \sum_{u \in \{0, 1\}^d} \Pr[\mathcal{H}(u) = \mathcal{H}(x)] \\ &\leq \left| \{u \in \{0, 1\}^d : \|u - x\|_1 \leq R\} \right| \\ &\quad + q \cdot \left| \{u \in \{0, 1\}^d : \|u - x\|_1 > R\} \right| \\ &= \sum_{k=0}^{\lfloor R \rfloor} \binom{d}{k} + q \cdot \sum_{k=\lfloor R \rfloor+1}^d \binom{d}{k} , \end{aligned}$$

which is the required inequality. $\square$

COROLLARY 2.3. *Assume that $R < \frac{d}{2}$. Then, using the notation of Lemma 2.2, we have that*

$$\mathbb{E} |\mathcal{H}^{-1}(\mathcal{H}(x))| \leq 2^d \left(q + e^{-\frac{1}{d}(\frac{d}{2} - R)^2} \right) .$$

PROOF. This follows from Lemma 2.2 and the standard estimate $\sum_{k \leq \frac{d}{2} - a} \binom{d}{k} \leq 2^d \cdot e^{-\frac{a^2}{d}}$. $\square$

LEMMA 2.4 (RANDOM WALK LEMMA). *Let r be an odd integer. Given $\emptyset \neq B \subseteq \{0, 1\}^d$, consider the random variable $Q_B \in \{0, 1\}^d$ defined as follows: Choose a point $z \in B$ uniformly at random, and perform r -steps of the standard random walk on the Hamming cube starting from z . The point thus obtained will be denoted Q_B . Then*

$$\Pr[Q_B \in B] \leq \left(\frac{|B|}{2^d} \right)^{\frac{e^{2r/d} - 1}{e^{2r/d} + 1}} .$$

PROOF. We begin by recalling some background and notation on Fourier analysis on the Hamming cube. Given

$S \subseteq \{1, \dots, d\}$, the Walsh function $W_S : \{0, 1\}^d \rightarrow \{-1, 1\}$ is defined by

$$W_S(u) = (-1)^{\sum_{j \in S} u_j}.$$

For $f : \{0, 1\}^d \rightarrow \mathbb{R}$ we set

$$\widehat{f}(S) = \frac{1}{2^d} \sum_{u \in \{0, 1\}^d} f(u) W_S(u),$$

so that f can be decomposed as follows:

$$f = \sum_{S \subseteq \{1, \dots, d\}} \widehat{f}(S) W_S.$$

For every $f, g : \{0, 1\}^d \rightarrow \mathbb{R}$ we write

$$\langle f, g \rangle = \frac{1}{2^d} \sum_{u \in \{0, 1\}^d} f(u) g(u).$$

By Parseval's identity,

$$\langle f, g \rangle = \sum_{S \subseteq \{1, \dots, d\}} \widehat{f}(S) \widehat{g}(S).$$

For $\varepsilon \in [0, 1]$ the Bonami-Beckner operator T_ε is defined as

$$T_\varepsilon f = \sum_{S \subseteq \{1, \dots, d\}} \varepsilon^{|S|} \widehat{f}(S) W_S.$$

The Bonami-Beckner inequality [3, 2] states that for every $f : \{0, 1\}^d \rightarrow \mathbb{R}$,

$$\begin{aligned} \sum_{S \subseteq \{1, \dots, d\}} \varepsilon^{2|S|} \widehat{f}(S)^2 &= \|T_\varepsilon f\|_2^2 \\ &= \frac{1}{2^d} \sum_{u \in \{0, 1\}^d} (T_\varepsilon f(u))^2 \\ &\leq \|f\|_{1+\varepsilon^2}^2 \\ &= \left(\frac{1}{2^d} \sum_{u \in \{0, 1\}^d} f(u)^{1+\varepsilon^2} \right)^{\frac{2}{1+\varepsilon^2}}. \end{aligned}$$

Specializing to the indicator of $B \subseteq \{0, 1\}^d$ we get that

$$\sum_{S \subseteq \{1, \dots, d\}} \varepsilon^{2|S|} \widehat{\mathbf{1}_B}(S)^2 \leq \left(\frac{|B|}{2^d} \right)^{\frac{2}{1+\varepsilon^2}}. \quad (3)$$

Now, let P be the transition matrix of the standard random walk on $\{0, 1\}^d$, i.e. $P_{uv} = 1/d$ if u and v differ in exactly one coordinate, $P_{uv} = 0$ otherwise. By a direct computation we have that for every $S \subseteq \{1, \dots, d\}$,

$$P W_S = \left(1 - \frac{2|S|}{d} \right) W_S,$$

i.e. W_S is an eigenvector of P with eigenvalue $1 - \frac{2|S|}{d}$. The probability that the random walk starting from a random

point in B ends up in B after r steps equals

$$\begin{aligned} \Pr[Q_B \in B] &= \frac{1}{|B|} \sum_{a, b \in B} (P^r)_{ab} \\ &= \frac{2^d}{|B|} \langle P^r \mathbf{1}_B, \mathbf{1}_B \rangle \\ &= \frac{2^d}{|B|} \sum_{S \subseteq \{1, \dots, d\}} \widehat{\mathbf{1}_B}(S)^2 \left(1 - \frac{2|S|}{d} \right)^r \\ &\leq \frac{2^d}{|B|} \sum_{\substack{S \subseteq \{1, \dots, d\} \\ |S| \leq d/2}} \widehat{\mathbf{1}_B}(S)^2 \left(1 - \frac{2|S|}{d} \right)^r, \end{aligned}$$

where we used the fact that r is odd (i.e. we dropped negative terms).

Thus, using (3) we see that

$$\begin{aligned} \Pr[Q_B \in B] &\leq \frac{2^d}{|B|} \sum_{S \subseteq \{1, \dots, d\}} \widehat{\mathbf{1}_B}(S)^2 \cdot e^{-2r|S|/d} \\ &\leq \frac{2^d}{|B|} \cdot \left(\frac{|B|}{2^d} \right)^{\frac{2}{1+e^{-2r/d}}} \\ &= \left(\frac{|B|}{2^d} \right)^{\frac{1-e^{-2r/d}}{1+e^{-2r/d}}}, \end{aligned}$$

completing the proof Lemma 2.4. $\square$

PROOF OF PROPOSITION 2.1. Assume that r is an odd integer and $R < \frac{d}{2}$. For $x \in \{0, 1\}^d$ let $W_r(x) \in \{0, 1\}^d$ be the random point obtained by preforming a random walk for r steps starting at x . Since $\|x - W_r(x)\|_1 \leq r$ we know that $\Pr[\mathcal{H}(W_r(x)) = \mathcal{H}(x)] \geq p$. Taking expectation with respect to the uniform probability measure on $\{0, 1\}^d$ we deduce that

$$\begin{aligned} p &\leq \mathbb{E}_{x \in \{0, 1\}^d} \Pr[\mathcal{H}(W_r(x)) = \mathcal{H}(x)] \\ &= \mathbb{E}_{\mathcal{H}} \Pr[x \in \{0, 1\}^d : W_r(x) \in \mathcal{H}^{-1}(\mathcal{H}(x))] \\ &= \mathbb{E}_{\mathcal{H}} \sum_{k \in \mathbb{N}} \Pr[x \in \{0, 1\}^d : \\ &\quad W_r(x) \in \mathcal{H}^{-1}(\mathcal{H}(x)) \wedge \mathcal{H}(x) = k] \\ &= \mathbb{E}_{\mathcal{H}} \sum_{k \in \mathbb{N}} \frac{|\mathcal{H}^{-1}(k)|}{2^d} \Pr[Q_{\mathcal{H}^{-1}(k)} \in \mathcal{H}^{-1}(k)] \\ &\leq \mathbb{E}_{\mathcal{H}} \sum_{k \in \mathbb{N}} \frac{|\mathcal{H}^{-1}(k)|}{2^d} \cdot \left(\frac{|\mathcal{H}^{-1}(k)|}{2^d} \right)^{\frac{e^{2r/d}-1}{e^{2r/d}+1}} \quad (4) \end{aligned}$$

$$\begin{aligned} &= \mathbb{E}_{\mathcal{H}} \mathbb{E}_{x \in \{0, 1\}^d} \left(\frac{|\mathcal{H}^{-1}(\mathcal{H}(x))|}{2^d} \right)^{\frac{e^{2r/d}-1}{e^{2r/d}+1}} \\ &\leq \mathbb{E}_{x \in \{0, 1\}^d} \left(\frac{\mathbb{E}_{\mathcal{H}} |\mathcal{H}^{-1}(\mathcal{H}(x))|}{2^d} \right)^{\frac{e^{2r/d}-1}{e^{2r/d}+1}} \quad (5) \end{aligned}$$

$$\leq \left(q + e^{-\frac{1}{d}(\frac{d}{2}-R)} \right)^{\frac{e^{2r/d}-1}{e^{2r/d}+1}}, \quad (6)$$

where in (4) we used Lemma 2.4, in (5) we used Jensen's inequality, and in (6) we used Corollary 2.3. $\square$

3. ACKNOWLEDGEMENTS

We are grateful to Piotr Indyk and Jirka Matoušek for helpful suggestions.

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